ENGINEERING IMPLEMENTATION REPORT

RAG & Vector Search

Implementation and Testing Report

Module	Retrieval-Augmented Generation and Vector Search
Assigned Roll Number	**25071A0567**
Endpoint	`POST /api/contributor-chat`
Repository	open-source-contributor-project
Status	Implemented; 69 automated tests passed; live Ollama and Atlas checks reported successful by the contributor.
Report Date	11 October 2026

1. Executive Summary

This work completes the project's existing Retrieval-Augmented Generation (RAG) path. Imported snapshot documents are split into searchable chunks, embedded through the configured Ollama model, and stored in MongoDB with the metadata required by Atlas Vector Search.

Contributor questions are checked against the user's project access and the project's current snapshot before retrieval. Responses are based on retrieved evidence, with citations built from real chunk file paths and the snapshot version. When no relevant evidence is available, the API returns **"Not found in snapshot."**

> **Verification note:** The contributor reports successful live verification of Ollama and Atlas. That live check was not independently repeated while preparing this report; the automated test suite uses mocks for external services.

2. Architecture & Workflow

Imported snapshot	->	**Overlapping chunks**	->	**Ollama embeddings**
MongoDB Chunk records	->	**Authorized Atlas search**	->	**Grounded answer + citations**

Technology stack: JavaScript, Node.js 20+, Express, Mongoose, MongoDB Atlas Vector Search, Ollama, and LangChain JS.

IMPLEMENTATION DETAILS

3. Snapshot Ingestion and Embedding

Chunking

The new RAG ingestion service divides each imported document into chunks of up to **1,200 characters** with **150 characters of overlap**. It prefers nearby paragraph and line breaks while keeping chunk sizes bounded. Each chunk retains its source file path and per-document chunk index.

Embedding generation

The service uses the existing environment configuration: `OLLAMA_URL` and `EMBED_MODEL`. The project default for the embedding model is `nomic-embed-text`. Each result is checked to ensure it has exactly 768 finite numeric values before it can be saved.

```
OllamaEmbeddings({ baseUrl: env.OLLAMA_URL, model: env.EMBED_MODEL })
embedDocuments(chunkTexts)
```

Chunk storage

Chunk records include the existing schema fields used by the search path:

- `projectId`, `snapshotId`, and `documentId`
- `visibility` copied from the project
- `filePath`, `chunkIndex`, and chunk `text`
- `embedding` with 768 dimensions

Safe snapshot activation

Embedding is completed before the route writes the new snapshot. The route writes the snapshot, documents, chunks, and imported issues before updating `currentSnapshotId`. If a database write fails, it attempts to remove records for that newly created snapshot and does not advance the project's current snapshot pointer.

> **Scope:** This adds embedding and chunk storage to snapshot import; it does not replace or rewrite the project's existing import architecture.

RETRIEVAL, ACCESS CONTROL & ANSWERS

4. Contributor Chat and Retrieval

Authorization and snapshot selection

The contributor-chat route requires authentication and applies the project's existing public/private access rules before looking up the current snapshot or running vector retrieval. It selects the snapshot referenced by the project's `currentSnapshotId` and verifies that the snapshot belongs to that project.

Public verified projects follow the contributor access path. Private or unverified projects retain the existing maintainer/moderator restrictions. Retrieval is constrained to the authorized project, its current snapshot, and its permitted visibility.

Atlas Vector Search configuration

Setting	Required definition
Collection / index	`chunks` / `chunks_vector_index`
Vector field	`embedding`, 768 dimensions, cosine similarity
Filter fields	`projectId`, `snapshotId`, `visibility`
Query behavior	Top-five chunks, with project, snapshot, and visibility filters

The checked-in index definition matches the vector-search stage's index name, vector path, dimensions, and filter fields. Live index presence is based on the contributor's report, not an Atlas query performed for this document.

Grounded answers and citations

- The answer model is instructed to use only retrieved repository evidence and return structured evidence numbers.
- The server maps selected evidence numbers back to the actual retrieved chunks.
- Citation file paths and snapshot versions are constructed by the server, not invented by the model.
- No retrieved or supporting evidence produces the fixed "Not found in snapshot." answer with no citations.

AUTOMATED VERIFICATION

5. Test Results

Command	`cd oss-onboarding/server && npm test`
Tests	**69 passed / 69 total**
Failures / skips	0 failures / 0 skipped
External services	Mocked in automated tests

The last full suite run completed successfully before this report was prepared. It was not rerun for report generation.

Coverage relevant to RAG

- Chunk size, overlap, paragraph boundaries, and per-document chunk numbering.
- Embedding count and 768-dimensional finite vector validation.
- Chunk metadata required by the Mongoose schema and vector filters.
- Successful snapshot import, Ollama configuration, and chunk storage call.
- Embedding and chunk-write failures do not activate an incomplete snapshot.
- Existing contributor-chat authentication, authorization-before-retrieval, vector filters, grounded citations, and empty-result behavior.

> **Live verification:** The contributor states that Ollama embedding generation and Atlas vector search were exercised successfully. Automated tests remain mock-based and are not a substitute for that live check.

6. Files Changed

File	Purpose
`server/src/services/rag/ingestionService.js`	Document chunking, Ollama embeddings, and vector validation.
`server/src/routes/projects.routes.js`	Integrates ingestion into snapshot import and protects current-snapshot activation.
`server/tests/projectsApi.test.js`	Import success, model configuration, chunk storage, and failure handling tests.
`server/tests/ragIngestion.test.js`	Chunk splitting, metadata, empty content, and embedding validation tests.

CONFIGURATION & HANDOFF

7. Environment and Remaining Limitations

- `EMBED_MODEL` defaults to `nomic-embed-text`; `OLLAMA_URL` defaults to `http://localhost:11434`. The selected Ollama service must have the model available.
- The server requires `MONGODB_URI`. The Replit development environment was previously checked and did not have this variable configured, so live verification could not be reproduced from that environment during this work.
- Atlas must have the `chunks_vector_index` configured on the `chunks` collection with the definition listed in this report.
- Do not use a production database for development verification. Configure credentials through the workspace's development secrets mechanism; never place them in source code or this report.

Known boundary

The server constrains answer generation to retrieved evidence and constructs citations from real chunks. As with any LLM, prompt-following alone cannot mathematically guarantee semantic entailment; applications should continue to review answer quality against the cited source text.

8. Conclusion

The RAG path now has a real snapshot-ingestion implementation connected to the existing contributor-chat retrieval flow. Automated verification passed all 69 tests. The contributor reports successful live Ollama and Atlas verification; this report clearly separates that confirmation from the mock-based automated tests and from checks independently performed in this workspace.

> **Delivery status:** Report prepared for roll number **25071A0567**. No project code was changed while preparing this document, and no commit, push, or deployment was performed.